CS 4412 — M3 Analysis Document

Retail Transaction Pattern Discovery
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Milestone: M3 Complete Implementation

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1. Implementation Status

The M3 implementation is now functionally complete and operates end-to-end. The notebook loads the retail transaction dataset, cleans and preprocesses the data, performs exploratory data analysis, constructs transaction baskets, mines association rules, compares rule-mining approaches, and exports results for reuse in the final report and repository.

Compared with M2, the notebook now includes analytical components that were previously missing, including summary statistics, visualizations, markdown section headers, interpretive discussion, and explicit justification of the analysis scope. These additions address earlier feedback that the analysis moved too quickly from loading data to association rule mining without first understanding the dataset.

The project now includes both Apriori and FP-Growth implementations, along with exploratory data analysis, making the workflow more robust and analytically complete.

2. Analytical Choices

The project focuses on United Kingdom transactions and a filtered subset of valid retail records. This decision was made to make market-basket mining computationally manageable while still preserving enough transactions for meaningful pattern discovery. However, this choice also introduces a limitation, since the resulting rules may reflect UK-specific purchasing behavior rather than broader global patterns.

Data cleaning steps included:

- Removing missing customer IDs
- Removing cancelled invoices
- Removing zero or negative quantities
- Removing zero or negative prices

These steps improved data quality and reduced noise in the mining process.

Apriori remains the core M2 technique because the primary discovery question focuses on identifying products frequently purchased together. For M3, the analysis expands beyond Apriori by adding FP-Growth to test whether co-purchase patterns remain consistent across algorithms.

This expansion strengthens the reliability of findings and demonstrates analytical improvement beyond a single mining approach.

3. Results

The association-rule mining stage produced several interpretable co-purchase relationships rather than only raw output tables. One of the strongest rules discovered shows that customers purchasing Strawberry Ceramic Trinket Box are highly likely to also purchase Sweetheart Ceramic Trinket Box. This rule exhibited high lift and confidence values, indicating a strong complementary relationship between decorative gift items.

Another strong association appeared between Wooden Frame Antique White and Wooden Picture Frame White Finish. This pattern suggests customers frequently purchase coordinated decorative items, indicating a preference for matching home décor products.

Additionally, the rule connecting Heart of Wicker Small and Heart of Wicker Large suggests that customers often purchase multiple sizes of the same product. This behavior indicates potential opportunities for size-based bundling and multi-item promotions.

The FP-Growth comparison confirmed that many of the strongest rules discovered by Apriori also appeared in FP-Growth results. This overlap strengthens confidence in the discovered purchasing patterns and suggests that the relationships are stable across different mining techniques.

Overall, the results demonstrate that association rule mining can successfully identify meaningful purchasing patterns in retail transactions. These findings provide actionable insights that could support recommendation systems, product bundling, and targeted marketing strategies.

4. Discovery Question Progress

The primary discovery question asked which products are frequently purchased together in UK retail transactions. This question is now being answered through multiple high-lift association rules identified in the analysis.

A second discovery question examined whether restricting analysis to UK transactions affects interpretation. Exploratory data analysis and discussion of scope limitations address this concern, acknowledging that findings represent scoped patterns rather than universal retail behavior.

A third discovery question examined whether multiple mining techniques produce consistent results. The comparison between Apriori and FP-Growth directly addresses this question and confirms that the strongest co-purchase patterns remain stable across algorithms.

These additions demonstrate that the project has progressed beyond simple rule generation and now includes deeper analytical interpretation.

5. M4 Plan

For M4, the technical pipeline does not require major modifications. The remaining work focuses on improving presentation and interpretation. The strongest rules and cluster findings will be written more clearly in business-focused language.

The final report will also include:

- Improved visualizations
- Clearer interpretation of findings
- Discussion of limitations
- Repository cleanup and organization

The final submission will emphasize clarity, presentation quality, and professional reporting of findings.